

1 Gain law

(bars denote normalized quantities: lengths by R , $\bar{a} = a/a_o$, $\bar{f} = a_o f$; $0 < \bar{a} < 1$; $e_\theta = \theta - \hat{\theta}$)

$$\bar{a}_{\text{true}}(\bar{w}) = 1 - \frac{9}{8\bar{w}} + \frac{1}{2\bar{w}^3} - \frac{57}{100\bar{w}^4} + \frac{1}{5\bar{w}^5} + \frac{7}{200\bar{w}^{11}} - \frac{1}{25\bar{w}^{12}}$$

$$\bar{a}'_{\text{true}}(\bar{w}) > 0 \quad \text{on} \quad \bar{w} > 1 \quad \implies \quad \bar{a}' = \frac{(1 - \bar{a})^2}{b_{\text{true}}(\bar{a})} \quad \text{exactly,} \quad b_{\text{true}} := \frac{(1 - \bar{a}_{\text{true}})^2}{\bar{a}'_{\text{true}}}$$

$$\begin{aligned} \bar{a}_{\text{true}}(1) = 0, \quad \bar{a}'_{\text{true}}(1) = 1 &\implies b_{\text{true}}(\bar{a} = 0) = 1 \\ 1 - \bar{a}_{\text{true}} = \frac{9}{8\bar{w}} + \mathcal{O}(\bar{w}^{-3}) &\implies b_{\text{true}}(\bar{a} \rightarrow 1) = \frac{9}{8} \end{aligned}$$

$$\begin{aligned} \bar{a}'(\bar{a}, b) &:= \frac{(1 - \bar{a})^2}{b} \\ a_w &= a_o \bar{a}_w, \quad a_o = \frac{\Delta t}{\gamma_N R} \end{aligned}$$

2 Jacobians

$$\begin{aligned} \frac{\partial \bar{a}'}{\partial \bar{a}} &= -\frac{2(1 - \bar{a})}{b} \\ \frac{\partial \bar{a}'}{\partial b} &= -\frac{(1 - \bar{a})^2}{b^2} = -\frac{\bar{a}'}{b} \end{aligned}$$

3 True dynamics

$$(\delta \bar{w}_j[k] := \delta \bar{w}[k - d + j - 1], \quad j = 1, \dots, d + 1, \quad d = 2)$$

$$\begin{aligned} \bar{w}[k] &= \bar{w}_d[k] - \delta \bar{w}_3[k] \\ \bar{w}_d[k + 1] &= \bar{w}_d[k] + \Delta \bar{w}_d[k] \\ \delta \bar{w}_1[k + 1] &= \delta \bar{w}_2[k] \\ \delta \bar{w}_2[k + 1] &= \delta \bar{w}_3[k] \\ \delta \bar{w}_3[k + 1] &= \lambda_c \delta \bar{w}_3[k] - F_{dw}[k] e_{\bar{a}_w}[k] - \varepsilon_w[k] \\ \bar{a}_w[k + 1] &= \bar{a}_w[k] + \bar{a}'_{\text{true}}[k] \left[\Delta \bar{w}_d[k] + (1 - \lambda_c) \delta \bar{w}_3[k] + F_{dw}[k] e_{\bar{a}_w}[k] + \varepsilon_w[k] \right] \\ b[k + 1] &= b[k] \end{aligned}$$

$$F_{dw}[k] = \bar{f}_{dw}[k] + (1 - \lambda_c) \sum_{i=1}^d \bar{f}_{dw}[k - i]$$

$$\varepsilon_w[k] = \bar{a}_w[k] \bar{f}_T[k] + (1 - \lambda_c) \sum_{i=1}^d \bar{a}_w[k - i] \bar{f}_T[k - i] + (1 - \lambda_c) n_w[k]$$

4 Process model

$$\mathbf{x} = [\delta\bar{w}_1, \delta\bar{w}_2, \delta\bar{w}_3, \bar{a}_w, b]^\top \quad (d = 2)$$

5 Estimator

$$\begin{aligned} \delta\hat{w}_1[k+1] &= \delta\hat{w}_2[k] \\ \delta\hat{w}_2[k+1] &= \delta\hat{w}_3[k] \\ \delta\hat{w}_3[k+1] &= \lambda_c \delta\hat{w}_3[k] \\ \hat{a}_w[k+1] &= \hat{a}_w[k] + \frac{(1 - \hat{a}_w[k])^2}{\hat{b}[k]} [\Delta\bar{w}_d[k] + (1 - \lambda_c) \delta\hat{w}_3[k]] \\ \hat{b}[k+1] &= \hat{b}[k] \end{aligned}$$

6 Error dynamics

(True – Estimator; coefficients at the estimate)

$$\begin{aligned} e_{\delta\bar{w}_1}[k+1] &= e_{\delta\bar{w}_2}[k] \\ e_{\delta\bar{w}_2}[k+1] &= e_{\delta\bar{w}_3}[k] \\ e_{\delta\bar{w}_3}[k+1] &= \lambda_c e_{\delta\bar{w}_3}[k] - F_{dw}[k] e_{\bar{a}_w}[k] - \varepsilon_w[k] \\ e_{\bar{a}_w}[k+1] &= \left[1 + \frac{(1 - \hat{a}_w)^2}{\hat{b}} F_{dw} - \frac{2(1 - \hat{a}_w)}{\hat{b}} (\Delta\bar{w}_d + (1 - \lambda_c) \delta\hat{w}_3) \right] e_{\bar{a}_w}[k] \\ &\quad + \frac{(1 - \lambda_c)(1 - \hat{a}_w)^2}{\hat{b}} e_{\delta\bar{w}_3}[k] - \frac{(1 - \hat{a}_w)^2}{\hat{b}^2} (\Delta\bar{w}_d + (1 - \lambda_c) \delta\hat{w}_3) e_b[k] \\ &\quad + \frac{(1 - \hat{a}_w)^2}{\hat{b}} \varepsilon_w[k] \\ e_b[k+1] &= e_b[k] \end{aligned}$$

$$F_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & \lambda_c & -F_{dw} & 0 \\ 0 & 0 & \frac{(1-\lambda_c)(1-\hat{a}_w)^2}{\hat{b}} & 1 + \frac{(1-\hat{a}_w)^2}{\hat{b}} F_{dw} & -\frac{(1-\hat{a}_w)^2}{\hat{b}^2} \\ 0 & 0 & 0 & -\frac{2(1-\hat{a}_w)}{\hat{b}} (\Delta \bar{w}_d + (1-\lambda_c) \delta \hat{w}_3) & \times (\Delta \bar{w}_d + (1-\lambda_c) \delta \hat{w}_3) \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{q}[k] = \left[0, 0, -\varepsilon_w, \frac{(1-\hat{a}_w)^2}{\hat{b}} \varepsilon_w, 0 \right]^\top$$

7 Measurement and innovation

$$\begin{aligned} \nabla_d \bar{w}_d[k] &= \bar{w}_d[k] - \bar{w}_d[k-d] \\ y_1[k] &= \delta \bar{w}_1[k] + n_w[k] \\ y_2[k] &= \bar{a}_w[k-d] + n_a[k] \end{aligned}$$

$$\bar{a}_w[k-d] = \bar{a}_w[k] - \frac{(1-\bar{a}_w[k])^2}{b} \nabla_d \bar{w}_d[k]$$

$$\begin{aligned} \frac{\partial y_2}{\partial \bar{a}_w} &= 1 + \frac{2(1-\hat{a}_w)}{\hat{b}} \nabla_d \bar{w}_d \\ \frac{\partial y_2}{\partial b} &= \frac{(1-\hat{a}_w)^2}{\hat{b}^2} \nabla_d \bar{w}_d \end{aligned}$$

$$H[k] = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 + \frac{2(1-\hat{a}_w)}{\hat{b}} \nabla_d \bar{w}_d & \frac{(1-\hat{a}_w)^2}{\hat{b}^2} \nabla_d \bar{w}_d \end{bmatrix}$$

$$K[k] = P^f H^\top (H P^f H^\top + R)^{-1}$$

$$\begin{aligned} e_{y_1}[k] &= e_{\delta \bar{w}_1}[k] + n_w[k] \\ e_{y_2}[k] &= \left[1 + \frac{2(1-\hat{a}_w)}{\hat{b}} \nabla_d \bar{w}_d \right] e_{\bar{a}_w}[k] + \frac{(1-\hat{a}_w)^2}{\hat{b}^2} \nabla_d \bar{w}_d e_b[k] + n_a[k] \end{aligned}$$

8 Process and measurement noise

$$q_3 = -\bar{a}_w \bar{f}_T$$

$$q_4 = -\frac{(1 - \hat{a}_w)^2}{\hat{b}} q_3$$

$$\gamma(\bar{w}) = \gamma_N c(\bar{w})$$

$$\sigma_{\bar{f}_T}^2 = a_o^2 \frac{4k_B T \gamma(\bar{w})}{\Delta t}$$

$$Q_{33} = \frac{4k_B T a_o}{R} \bar{a}_w$$

$$Q_{34} = Q_{43} = -\frac{(1 - \hat{a}_w)^2}{\hat{b}} Q_{33}$$

$$Q_{44} = \frac{(1 - \hat{a}_w)^4}{\hat{b}^2} Q_{33}$$

$$Q_{55} = 0$$

$$R = \text{diag}(R_1, R_2)$$

$$R_1 = \sigma_{n_w}^2$$

$$R_2 = K_{\text{var}} I F_{\text{var}} (\bar{a}_w + \xi)^2 + d Q_{44}$$

$$K_{\text{var}} = \frac{2a_{\text{cov}}}{2 - a_{\text{cov}}}$$

$$\sigma_{\delta \bar{w}_r}^2 = C_{\text{dpmr}} \frac{4k_B T a_o}{R} (\bar{a}_w + \xi)$$

$$\xi = \frac{C_n \sigma_{n_w}^2}{C_{\text{dpmr}} (4k_B T a_o / R)}$$

Seed and prior

$$\bar{a}(\bar{w}) = 1 - \frac{b}{\bar{w} - \bar{w}_s}$$

$$\left. \frac{\partial \bar{a}}{\partial b} \right|_{\bar{w}} = -\frac{1}{\bar{w} - \bar{w}_s} = -\frac{1 - \bar{a}}{b}$$

$$\left. \frac{\partial \bar{a}}{\partial \bar{w}_s} \right|_{\bar{w}} = -\bar{a}'$$

$$\hat{b}[0] = \frac{1}{2} \left[\inf_{\bar{w} \in \mathcal{E}} b_{\text{true}} + \sup_{\bar{w} \in \mathcal{E}} b_{\text{true}} \right]$$

$$P_{55}[0] = P_{bb} = \left[\frac{1}{2} \left(\sup_{\bar{w} \in \mathcal{E}} b_{\text{true}} - \inf_{\bar{w} \in \mathcal{E}} b_{\text{true}} \right) \right]^2$$

$$\hat{\bar{a}}_w[0] = 1 - \frac{\hat{b}[0]}{\bar{w}[0] - \hat{\bar{w}}_s}$$

$$P_{44}[0] = (\hat{a}'[0])^2 P_{\bar{w}_s} + \left(\frac{1 - \hat{a}_w[0]}{\hat{b}[0]} \right)^2 P_{bb} + \text{floor}_a^2$$

$$P_{45}[0] = - \frac{1 - \hat{a}_w[0]}{\hat{b}[0]} P_{bb}$$

$$e_{\bar{a}}(\bar{w})|_{\bar{w}_s} = -\bar{a}'(\bar{w}) e_{\bar{w}_s} \propto \frac{b}{(\bar{w} - \bar{w}_s)^2}$$

$$e_{\bar{a}}(\bar{w})|_b = -\frac{e_b}{\bar{w} - \bar{w}_s}$$





